

Sanjay Dangi

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SUMMARY

Mechanical Engineer with strong proficiency in SolidWorks and AutoCAD, Geometric Dimensioning & Tolerancing (GD&T) and Bill of Material (BOM) management. Skilled in data handling, statistical and uncertainty analysis, and thermo-fluid CFD simulations. Applied FEA-based stress, fatigue, and failure analysis during design validation to optimize material selection and minimize manufacturing costs with hands-on exposure to industrial automation and PLC (ABB) based process control for production optimization.

SKILLS

- **Manufacturing Experience:** Production line operations in fertilizer and food manufacturing environments, including process monitoring, quality control (QC), SOP compliance, troubleshooting, and lean manufacturing practices.
- **Design Skills:** Product innovation, Design Engineering, Product Testing and validating, CAD, CAM, Drafting, Molding, Reverse Engineering, Top-down Modeling, Assembly Drawings, BOM, GD&T, Stress Analysis, Project Management, Technical Writing and Presentation
- **Design Software and Programming Language:** SolidWorks, AutoCAD, Creo, Python
- **Engineering Tools:** ANSYS Workbench, COMSOL Multiphysics, MATLAB, CNC Simulator, PLC, SCADA
- **Office & Productivity Tools:** MS PowerPoint, MS Excel, MS SharePoint, Office 365, LaTeX, Github

WORK EXPERIENCE

Graduate Research Assistant - NIU

Aug 2023 – Dec 2025

- Designed, fabricated, and optimized an electrohydrodynamic (EHD) vortex flow on high-voltage field for dust particle confinement using SolidWorks and CNC machining, water jet cutting, 3D printing, drilling.
- Conducted advanced particle movement experiments using particle image velocimetry (PIV) with high-energy laser technology to analyze vortex dynamics to improve system performance.
- Designed a perforated flow diffuser and suction system in SolidWorks, accounting for pressure drop, fabrication constraints, and mechanical integration within an existing system.
- Developed and implemented operational protocols for ESP testing, particle size and number measurement, and controlled sampling, ensuring accurate and repeatable data collection.
- Processed and quantified complex datasets with MATLAB, translating experimental results for system optimization and presenting findings at professional conferences.

Graduate Teaching Assistant - NIU

Aug 2024 – Dec 2025

- Mentored **Engineering Graphics** with 79 undergraduates, strengthening product design skills using SolidWorks.
- Assisted in teaching **Fluid Mechanics** with 50 undergraduate students, clarifying core concepts such as dimensional analysis, conservation laws, and flow behavior, while supporting numerical problem-solving.
- Facilitated student understandings with hands-on exercises reinforcing practical design principles.
- Coordinated accreditation-related tasks, improving workflow efficiency and supporting program quality initiatives.
- Taught and guided students at **Rockford School (RESA)** in 3D design and 3D printing using TinkerCAD, developing hands-on workshops to reinforce practical engineering and CAD skills.

Mechanical Engineering - Biogas and Fertilizer Production Plant, Nepal

May 2019 – June 2022

- Supervised and maintained mechanical systems for biogas generation, granular and liquid fertilizer production, including crusher, conveyor belt system, double rotary drum granulator, pumps, mixers, compressors, sensors, and piping networks, ensuring uninterrupted plant operations.
- Performed preventive and corrective maintenance on mechanical equipment, troubleshooting failures and implementing solutions to reduce downtime and improve system efficiency.
- Assisted in the design, assembly, and installation of process equipment, including reactors, digesters, decantation (solid/liquid separation) and material handling systems, applying mechanical design principles.
- Improved production efficiency by analyzing workflow and cycle times, implementing PLC updates, and optimizing human-machine SCADA interfaces for seamless industrial operations.
- Monitored production processes, analyzed operational data, and recommended improvements to enhance throughput, reduce energy consumption, and ensure consistent product quality to meet customer expectations.
- Coordinated cross-functional teams during plant upgrades and equipment commissioning, integrating mechanical, electrical, and control systems for seamless operations.

Industrial Internships at Toyota and Chaudhary Groups (CG) Foods

Jan 2018 – Aug 2018

- Studied and analyzed mechanical systems in automotive and noodle production industry, including conveyors, mixers, boilers, tray dryers and packaging units, identifying operational inefficiencies and potential failures.
- Applied manufacturing and engineering principles to improve process understanding, linking thermodynamics, fluid flow, and heat transfer concepts to real-world production.
- Gained practical exposure to Lean Manufacturing, standardized work, and process optimization techniques to enhance production efficiency and quality control.
- Observed industrial operations to assess equipment reliability, safety, and workflow, translating theoretical knowledge into actionable insights for manufacturing processes.

EDUCATION

Master of Science: Mechanical Engineering - NIU, Dekalb, IL

Aug 2023 – Dec 2025

GPA: 3.91/4.00 — Graduated in December 2025

Relevant Coursework:

Manufacturing Processes and Production Systems

Computer-Aided Design and Manufacturing (CAD/CAM) for tooling and fixtures

Design for Manufacturing & Assembly (DFMA)

Geometric Dimensioning & Tolerancing (GD&T) for quality control

Finite Element Analysis (FEA) for structural integrity and failure prevention

Rapid Prototyping and 3D Printing for production validation

Industrial Thermodynamics and Heat Transfer for process efficiency

Fluid Mechanics for material transport and flow systems

Energy Systems and Process Engineering

Projects: CAD and CFD Modeling, NIU

Aug 2023 – Aug 2025

- Designed a tile dispensing system using SolidWorks with detailed assemblies, technical drawings, GD&T, and BOM.
- Designed and assembled pick-and-place gripper system with custom end effector and pneumatic actuation, sourcing standard components from McMaster-Carr ensuring reliable, repeatable motion for automated product handling.
- Conducted stress and failure analysis using ANSYS, identifying the fillets of the lock component as the most critical failure point.
- Modeled heat-driven cavity flow, fluid flow simulation around the obstacle and temperature distribution during laser cutting using ANSYS, COMSOL and Python to analyze natural convection and thermal performance in enclosed systems.